

MARINE RED ALGAE: A NATURAL DEFENCE TO ALLEVIATE FOOD SPOILAGE SCIENCES 2026 RESEARCH SYMPOSIUM

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INTRODUCTION

Food spoilage is a major contributor to global food waste, causing substantial economic losses and food insecurity. Approximately one-third of food produced globally is lost or wasted annually, with microbial contamination contributing significantly throughout the food supply chain [1]. Synthetic preservatives are widely used to control microbial growth; however, increasing health and environmental concerns and consumer demand for clean-label foods have driven interest in natural alternatives. Marine red algae (Rhodophyta) are rich sources of diverse bioactive metabolites, including halogenated terpenes, alkaloids, carotenoids, phenolics, and sulphated polysaccharides. Despite their recognised bioactivities, their potential as natural agents to alleviate food spoilage remains largely underexplored. This study investigated the antimicrobial, antioxidant, cytotoxicity and chemical composition of selected macro red algae species *Callophyllis rangiferina* (CR), *Laurencia heteroclada* (LH) and *Laurencia filiformis* (LF).

METHODS

Crude extracts were prepared by solvent extraction [2]. Then evaluated antimicrobial activity against major spoilage microorganisms, foodborne and fungi using disc diffusion and broth microdilution assays. Antioxidant activity was assessed using the ABTS free radical scavenging assay, while total polyphenols (TPC) and flavonoids (TFC) were determined using the Folin–Ciocalteu and aluminium complexation methods, respectively. Comprehensive chemical profiling was performed using gas chromatography–time-of-flight mass spectrometry (GC–TOF–MS) and liquid chromatography–mass spectrometry (LC–MS), coupled with feature-based molecular networking (FBMN) to identify bioactive metabolites present in the algal extracts. Cytotoxicity was assessed in Caco-2 cells using the MTT cell viability assay.

RESULTS AND DISCUSSION

All extracts exhibited antimicrobial activity against selected Gram-positive and Gram-negative bacteria and fungi. CR showed potent broad-spectrum activity with lower MIC/MFC values (Table 1) and the highest antioxidant activity, TPC, and TFC. Both *Laurencia* spp. showed comparable antimicrobial and antioxidant activities. GC–TOF–MS revealed characteristic isotopic patterns of halogenated compounds, while LC–MS/FBMN identified diverse bioactive metabolite clusters, including terpenes, alkaloids, carotenoids, sterols, phenolic glycosides, and halogenated metabolites, potentially contributing to the observed antimicrobial and antioxidant activities. Cell viability decreased with increasing extract concentration, with *Callophyllis* sp. showing the lowest cytotoxicity (Figure 1).

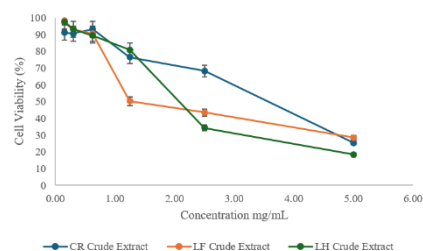


Figure 1 Effect of marine red algal extracts on Caco-2 cell viability after 24 h treatment (MTT assay)

CONCLUSIONS

These findings suggest that the three marine red algal species have promising potential for natural food preservation applications, requiring further testing in food model systems.

ACKNOWLEDGEMENTS

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REFERENCES

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Table 1: Minimum inhibitory/fungicidal concentration (MIC/MFC) ranges (mg/mL) of CR, LH and LF crude extracts.

	CR	LH	LF
Food Spoilage Microorganisms	0.039–0.625	0.078–2.500	0.039–1.250
Foodborne Pathogens	0.078–0.313	0.625–5.000	0.039–0.625
Fungi & Yeast	0.005–0.078	2.500–5.000	1.250–5.000